UMM AL-QURA UNIVERSITY

World War II Economic Data Warehouse

Our project is a Data Warehouse that integrates historical economic data from World War II, including gold prices, oil prices, and U.S dollar fluctuations. The goal is to analyze economic trends during the war and understand the relationships between these indicators.

DS22O2 – Final Project Presentation

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Lab1 : Data Sources

Project Scenario

Our project is a Data Warehouse that integrates historical economic data from World War II, including gold prices, oil prices, and U.S. dollar fluctuations. The goal is to analyze economic trends during the war and understand the relationships between these indicators.

Source name	Type	Format	Records	Notes
Monthly gold price on world	structured	csv	518	From Kaggle
Exchange rates	structured	csv	12175	From Kaggle
Oil prices	structured	csv	155	From Wikimedia

* Monthly Gold Price on World: Historical monthly gold prices used for trend analysis.
* Exchange Rates: Historical currency exchange rates used for economic analysis.
* Oil Prices: Historical oil prices used to study market fluctuations.

Lab2: Structured vs. Unstructured Data

Target Feature	Data Type	Structural Category	Source Context
Report_Text	Text	UNSTRUCTURED	Free-form narrative report blocks
Year	Integer	UNSTRUCTURED	Embedded chronologically inside paragraphs
Inflation_Rate	Float	UNSTRUCTURED	Described textually as percentage metrics
Gold_Reserve_Status	String	UNSTRUCTURED	Qualitative description of reserve conditions
Oil_Import_Status	String	UNSTRUCTURED	Narrative context regarding import levels

Transformation Methodology

1. Text Processing Pipeline

Programmatically ingested raw PDF financial reports and executed text segmentation to isolate data blocks chronologically per year[span_1](start_span)[span_1](end_span).

2. Feature Extraction Logic

Applied

Regex Pattern Matching

to convert narrative numbers into a structured column, and used

Keyword Detection

to parse qualitative descriptions[span_2](start_span)[span_2](end_span).

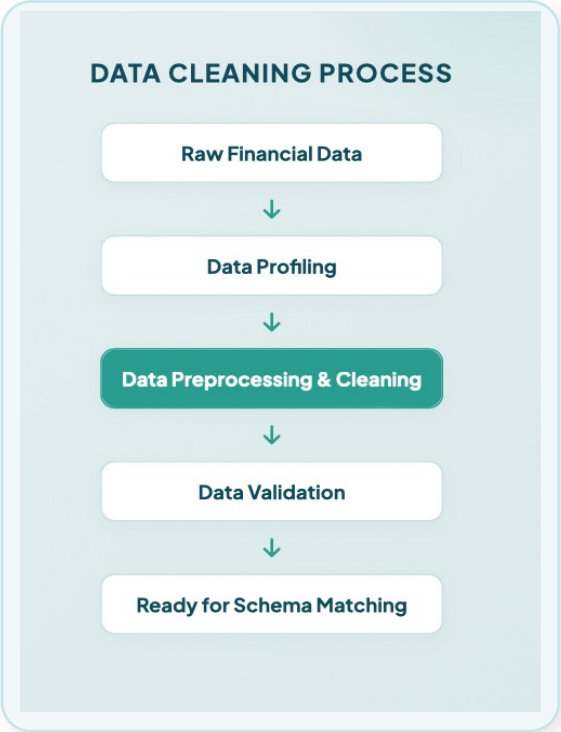
3. Target Schema Readiness

Synthesized all processed features into a clean, structured tabular DataFrame, while engineering a derived integration attribute (`War_Context`)[span_3](start_span)[span_3](end_span).

Lab3: Data Quality Assessment & Preprocessing

Data Quality Assessment & Cleaning

IDENTIFIED ISSUE	CLEANING ACTION / APPLIED RULE
Temporal Grain Mismatch	**Aggregation** `Mean(Daily_Price)` by Year
High Precision Decimals	**Formatting** `Round(Value, 2)` decimals
Weekend Market Closure	**Imputation** Forward Fill `(LOCF)`
Inconsistent Currencies	**Standardization** Map to standard ISO-3 Codes
Redundant Columns	**Dimensionality** Drop `Entity` constant values



Lab 4 : Schema Matching

In this lab :

This lab prepares schemas for mapping and uses schema matching to identify corresponding elements between different schemas for accurate data integration.

Source A Attribute	Source B Attribute	Match type	confidence	justification
Date	**Series Description**	Partial Match	Medium	**The names are different and the semantics are different. In the gold source, the time is monthly, while in the dollar source, the time is daily, but the structure is the same**
US dollar (USD)	**UNITED KINGDOM -- SPOT EXCHANGE RATE, US$/POUND (1/RXI_N.B.UK)**	No Match	Low	There is no match in semantics, structure, and name
Euro (EUR)	**SPOT EXCHANGE RATE - EURO AREA**	No Match	Low	There is no match in semantics, structure, and name
Japanese yen (JPY)	**JAPAN -- SPOT EXCHANGE RATE, YEN/US$**	No Match	Low	There is no match in semantics, structure, and name
Chinese renmimbi (CNY)	**CHINA -- SPOT EXCHANGE RATE, YUAN/US$ P.R.**	No Match	Low	There is no match in semantics, structure, and name

Lab5 : Schema Mapping

In this Lab:

We analyzed both schemas and noticed that Source A stores data on a daily basis, while Source B stores data monthly.
To make them compatible, we created mapping rules.
First, we extracted the month and year from each date.
Then, we grouped all daily records belonging to the same month and aggregated their values to produce monthly data that matches the target schema.

Source attribute	Target attribute	Mapping type	Short description
Date	Series Description	N Mapping:1	Daily dates in Source A are aggregated into monthly data in Source B
Date	Series Description	Conditional Mapping	The date is transformed based on a condition (grouping daily data into months)

Rule	Rule expression	Explanation
1	month = EXTRACT(MONTH FROM Date)	Extract the month from the daily date.
2	year = EXTRACT(YEAR FROM Date)	Extract the year to align with monthly data.
3	monthly_value = AGGREGATE(daily_values)	aggregation from daily to monthly granularity

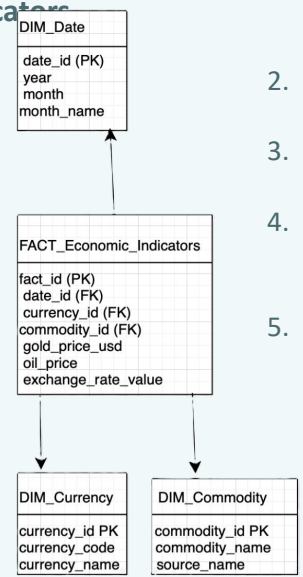
Lap 6 : star Schema + Data warehouse + ETL

Warehouse Design

Fact Table:
FACT_Economic_Indicators

Dimension Tables:
DIM_Date
DIM_Currency
DIM_Commodity

Fact Table Measures:
gold_price_usd
oil_price
exchange_rate_value.

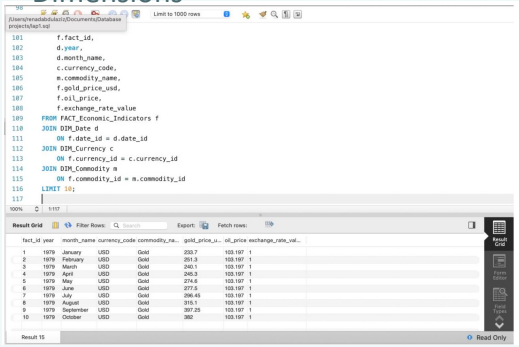


ETL Process

1. Extract data from gold, oil, and exchange rate CSV files.
2. Clean and standardize values.
3. Load dimension tables first.
4. Load fact table after dimensions to maintain foreign key relationships
5. Foreign keys `date_id`, `currency_id`, and `commodity_id` connect the fact table with dimension tables.

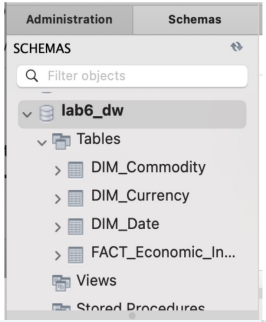
SQL insert codes

Joined Fact Table with Dimensions

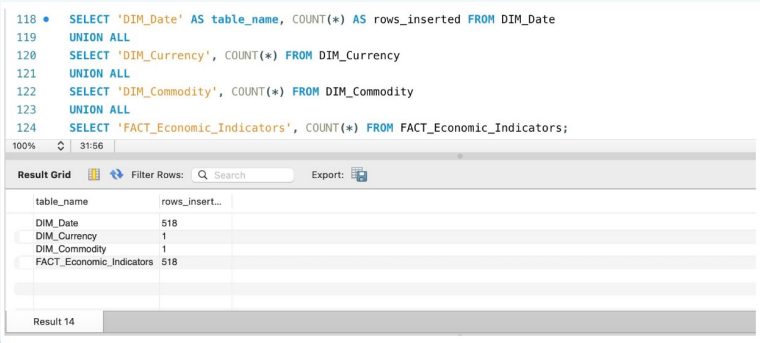


```
SELECT
    f.fact_id,
    d.year,
    d.month_name,
    c.currency_code,
    m.commodity_name,
    f.gold_price_usd,
    f.oil_price,
    f.exchange_rate_value
FROM FACT_Economic_Indicators f
JOIN DIM_Date d
    ON f.date_id = d.date_id
JOIN DIM_Currency c
    ON f.currency_id = c.currency_id
JOIN DIM_Commodity m
    ON f.commodity_id = m.commodity_id
LIMIT 10;
```

Created Warehouse Tables in MySQ



Rows Inserted Validation



```
SELECT 'DIM_Date' AS table_name, COUNT(*) AS rows_inserted FROM DIM_Date
UNION ALL
SELECT 'DIM_Currency', COUNT(*) FROM DIM_Currency
UNION ALL
SELECT 'DIM_Commodity', COUNT(*) FROM DIM_Commodity
UNION ALL
SELECT 'FACT_Economic_Indicators', COUNT(*) FROM FACT_Economic_Indicators;
```

Result Grid | Filter Rows: | Export:

table_name	rows_insert...
DIM_Date	518
DIM_Currency	1
DIM_Commodity	1
FACT_Economic_Indicators	518

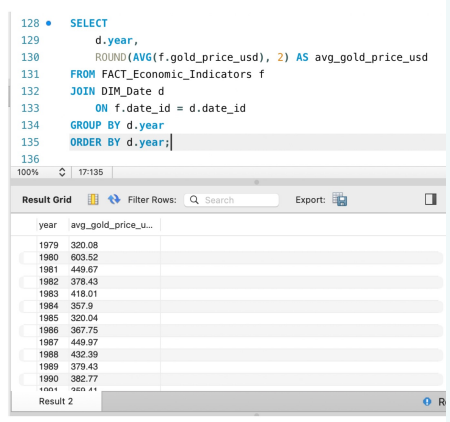
Result 14

Lab 7 : Strategic Questions + OLAP Analysis

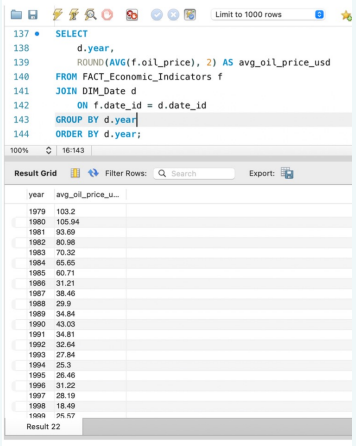
Summary table

Query#	Question	Tool	Result Summary
1	Which years recorded the highest average gold prices?	MySQL Workbench Bar Chart +Line Chart	Highest in 2022 (1852.55 USD), followed by 2021 and 2020, showing steady growth from 1979.
2	How did the average oil price change across years?	MySQL Workbench Bar Chart +Line Chart	Fluctuated: peak in 2011 (117.23 USD), lowest in 1998 (18.49 USD), high in late 1970s and early 1980s.
3	What is the relationship between gold prices and exchange rates across years?	MySQL Workbench Multi-Axis Line Chart	Gold rose over years; exchange rate constant at 1 USD, so gold per exchange unit equals gold price.
4	Which months consistently show the highest average oil prices across all years?	MySQL Workbench Bar Chart	Nearly constant (~58.01 USD), with January & February slightly lower (57.88 USD); no clear seasonal pattern.

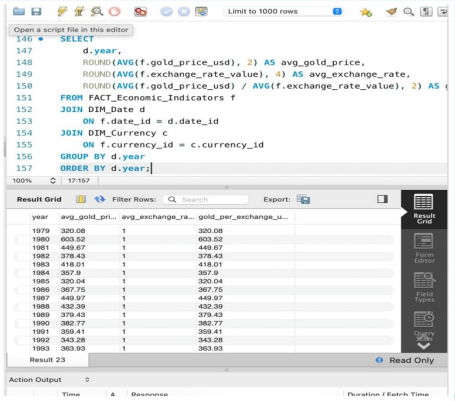
Q1



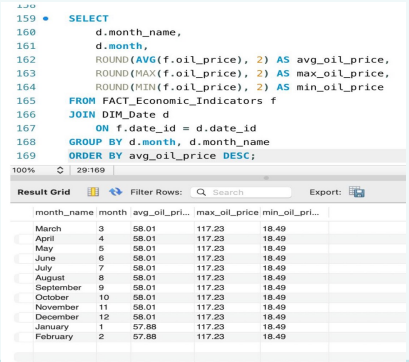
Q2



Q3



Q4



Key insights + Visual Dashboard

Key Insights:

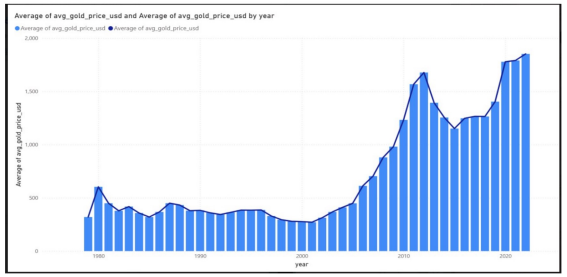
1. Gold prices showed a strong long-term increase, reaching the highest average value in 2022 at 1852.55 USD.

2. Oil prices fluctuated significantly across years, with the highest average in 2011 and the lowest in 1998.

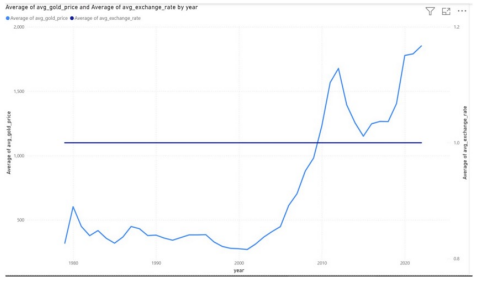
3. The exchange rate value used in the warehouse remained constant at 1 USD, so the gold value per exchange unit matched the gold price.

4. Monthly oil prices were almost stable across months, showing no strong seasonal pattern.

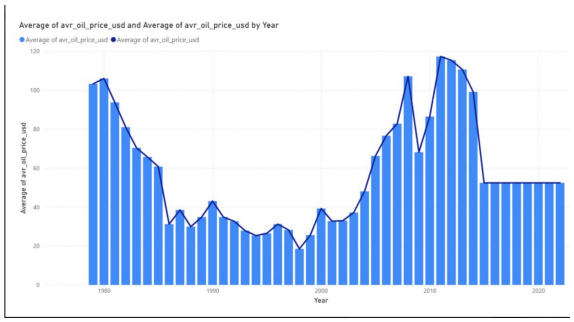
1



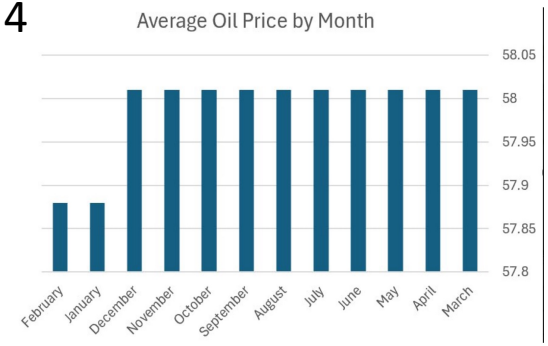
3



2



4



Final Recommendations + Lessons Learned

Recommendations

- Standardize dates and currency codes.
- Aggregate datasets to the same time level.
- Keep measures in fact table, attributes in dimension tables.
- Verify rows after each ETL step.
- Use multiple charts for analysis.

Lessons Learned

- Data cleaning and mapping are crucial.
- Star schema makes analysis easier.
- Load dimensions before fact table.
- Clear visuals improve understanding.